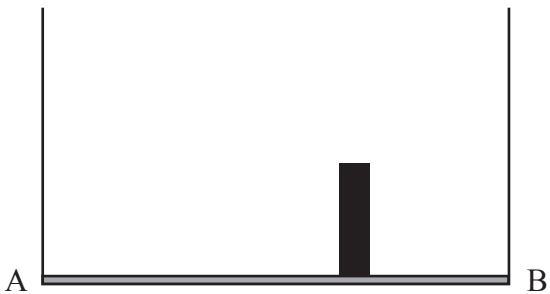
- The acceleration due to gravity is denoted by $g\,\text{m}\,\text{s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

6 A shelf consists of a horizontal uniform plank AB of length 0.8 m and mass 5 kg with light inextensible vertical strings attached at each end. A stack of bricks each of mass 2.3 kg is placed on the plank as shown in the diagram.



(a) Explain the meaning of each of the following modelling assumptions.

- The stack of bricks is modelled as a particle.
- The plank is modelled as uniform. **[2]**

Either of the strings will break if the tension exceeds 75 N.

(b) Find the greatest number of bricks that can be placed at the centre of the plank without breaking the strings. **[2]**

(c) Find an expression for the moment about A of the weight of a stack of n bricks when the stack is at a distance of x m from A. State the units for your answer. **[2]**

(d) Calculate the greatest distance from A that the largest stack of bricks can be placed without a string breaking. **[3]**

6(a)

6(b)

6(c)

6(d)